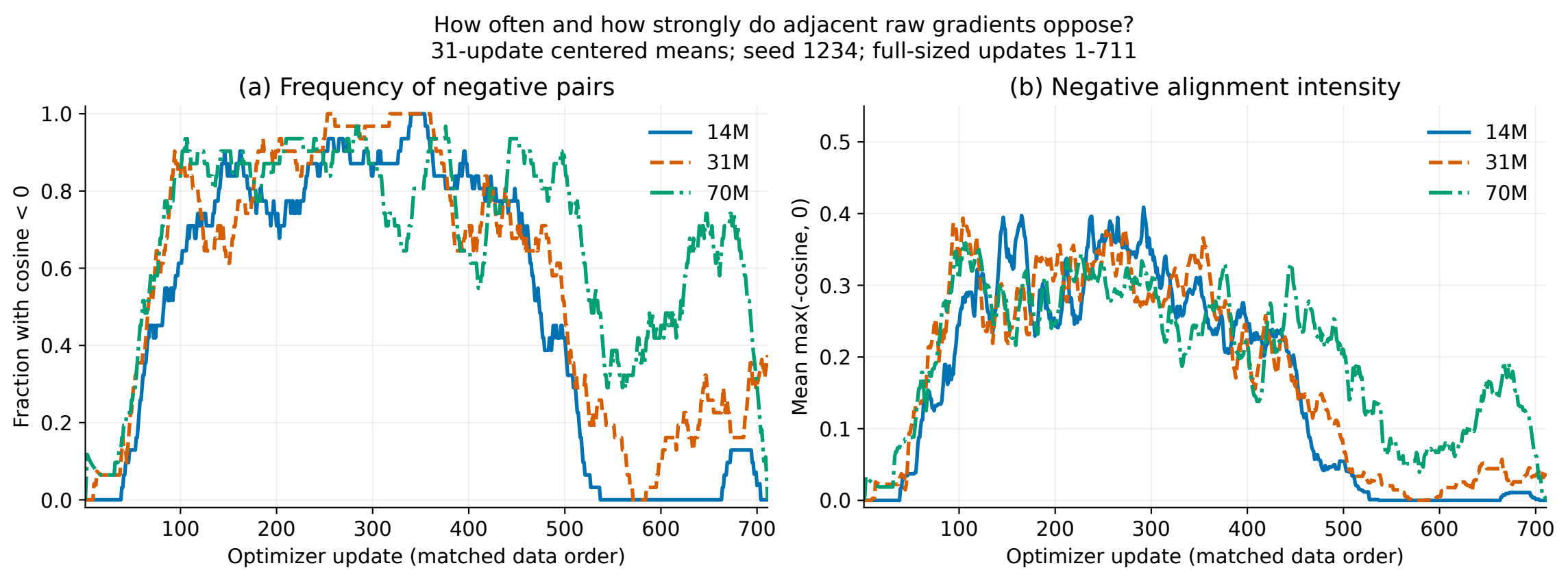

How often and how strongly do adjacent raw gradients oppose?
31-update centered means; seed 1234; full-sized updates 1-711
(a) Frequency of negative pairs
Fraction with cosine < 0
Optimizer update (matched data order)
(b) Negative alignment intensity
Mean max(-cosine, 0)
Optimizer update (matched data order)
14M
31M
70M